

Human Metapneumovirus (HMPV) Metabolic Model Analysis Report

Viral Biomass Objective Function (VBOF) Construction and Antiviral Target Identification

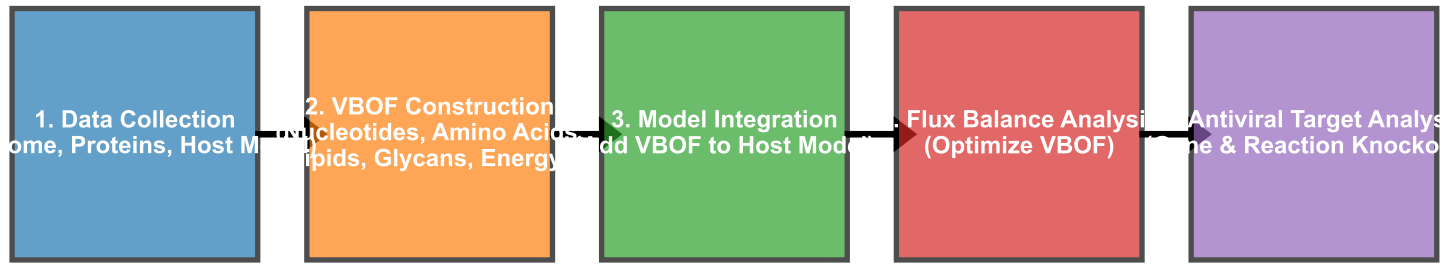
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HMPV Metabolic Modeling Project

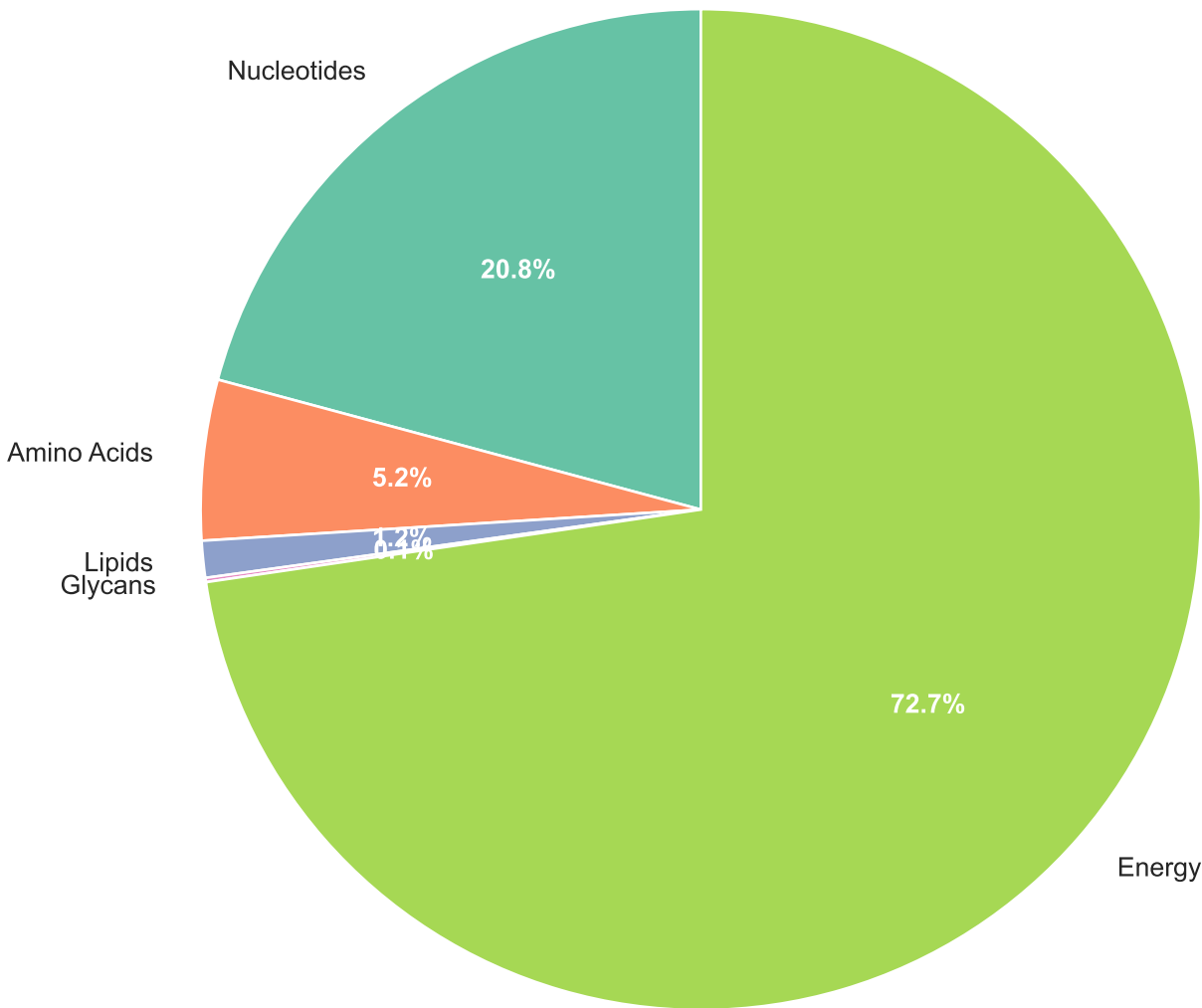
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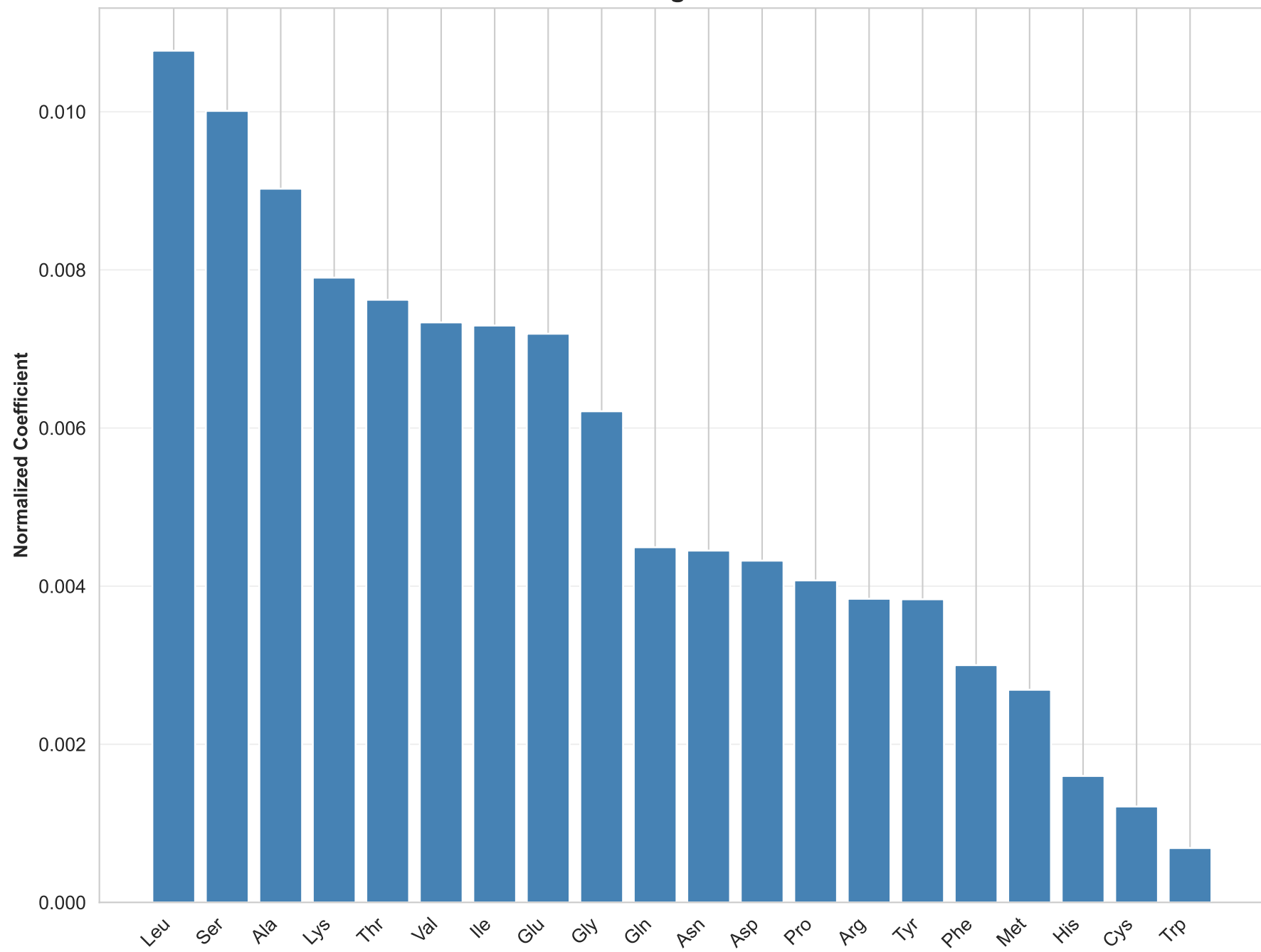
HMPV Metabolic Modeling Workflow



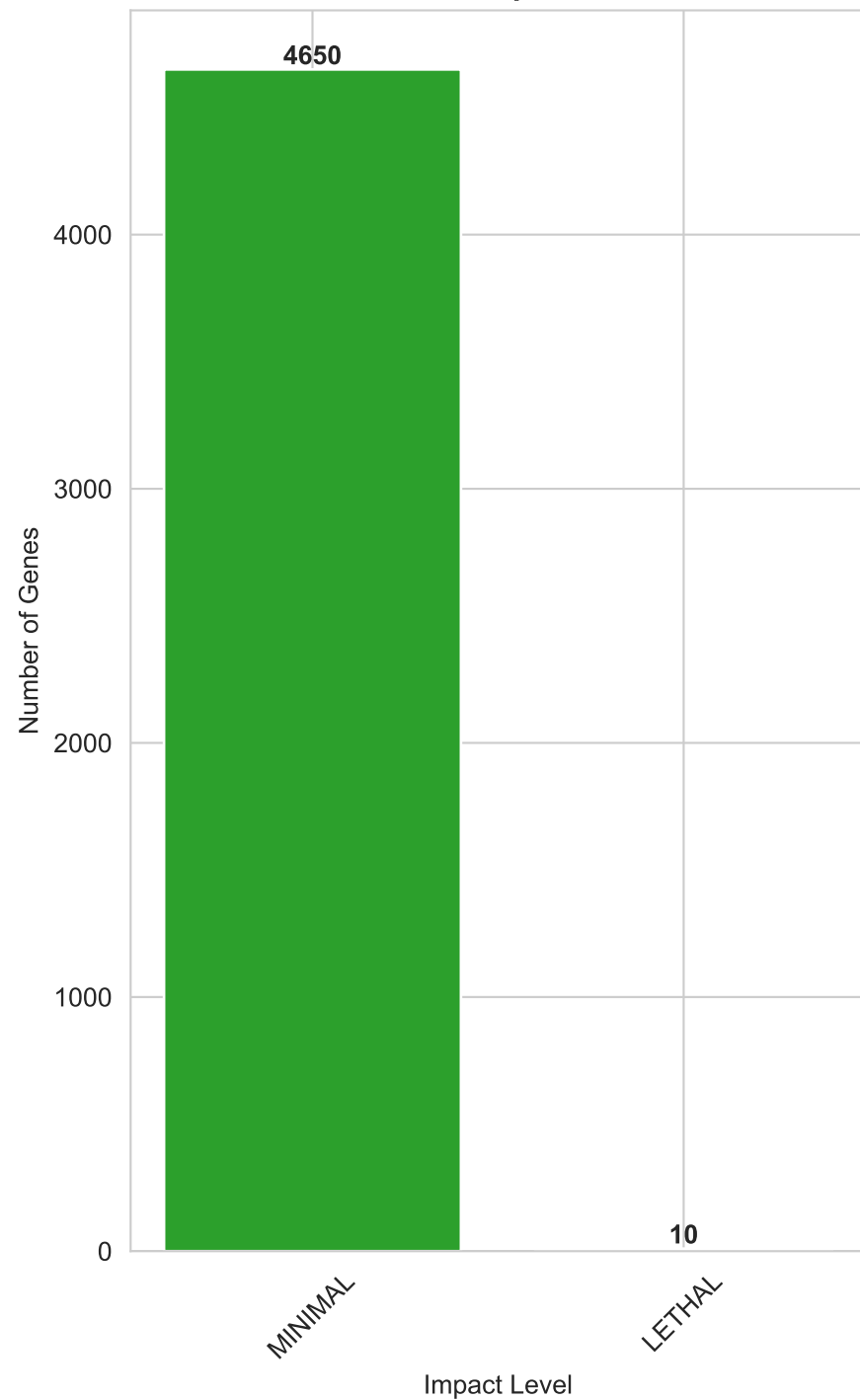
HMPV VBOF Composition by Component Type



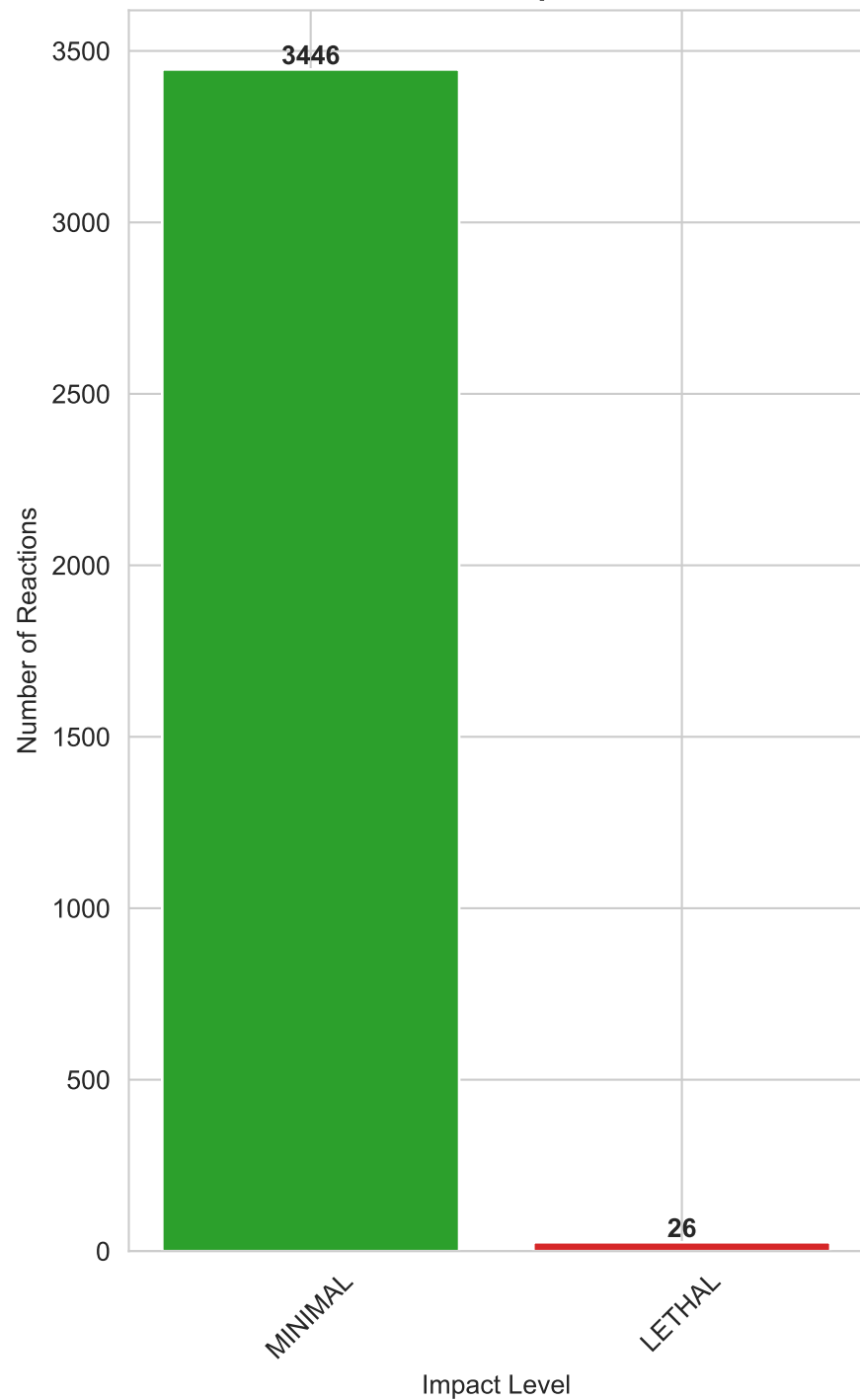
Amino Acid Usage in HMPV VBOF



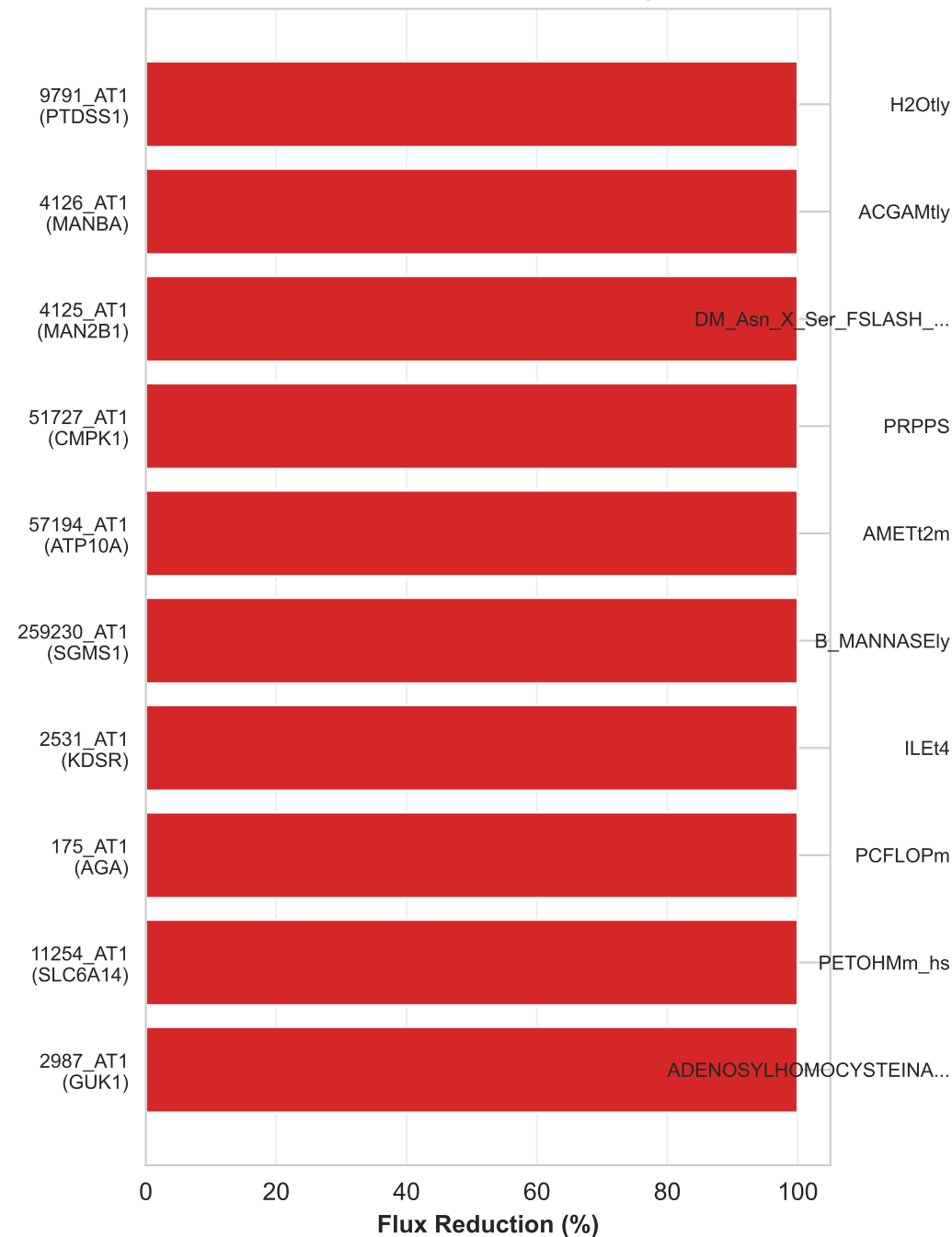
Gene Knockout Impact Distribution



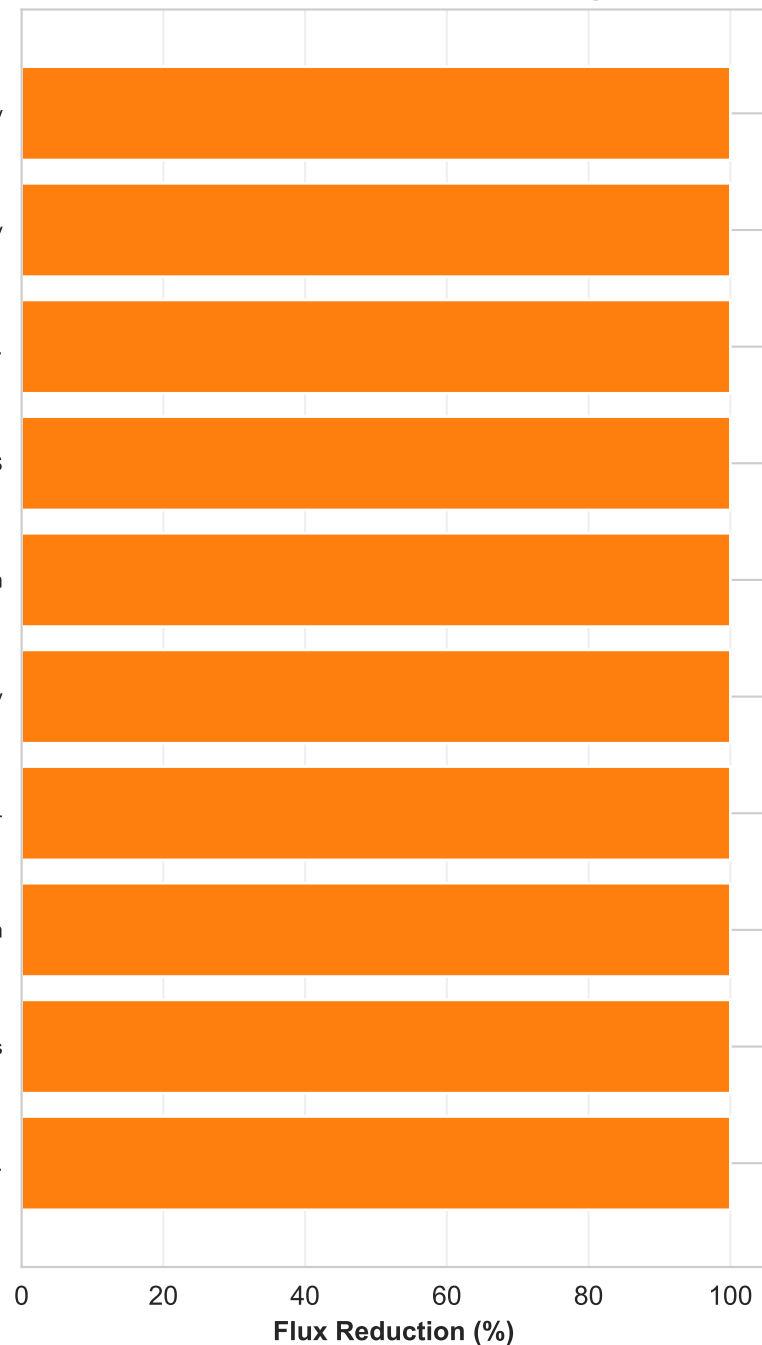
Reaction Knockout Impact Distribution



Top 10 Essential Gene Targets



Top 10 Essential Reaction Targets



Top 10 Essential Gene Targets for HMPV Production

Gene ID	Gene Name	Flux Reduction	Impact
9791_AT1	PTDSS1	100.0%	LETHAL
4126_AT1	MANBA	100.0%	LETHAL
4125_AT1	MAN2B1	100.0%	LETHAL
51727_AT1	CMPK1	100.0%	LETHAL
57194_AT1	ATP10A	100.0%	LETHAL
259230_AT1	SGMS1	100.0%	LETHAL
2531_AT1	KDSR	100.0%	LETHAL
175_AT1	AGA	100.0%	LETHAL
11254_AT1	SLC6A14	100.0%	LETHAL
2987_AT1	GUK1	100.0%	LETHAL

Top 10 Essential Reaction Targets for HMPV Production

Reaction ID	Reaction Name	Flux Reduction	Subsystem
H2Otly	H2O transport, lysosomal	100.0%	Unknown
ACGAMtly	N-acetyl-glucosamine lysosomal efflux	100.0%	Unknown
DM_Asn_X_Ser_FSLASH_Thr_LPAREN...	DM Asn-X-Ser/Thr(ly)	100.0%	Unknown
PRPPS	phosphoribosylpyrophosphate synthetase	100.0%	Unknown
AMETt2m	S-Adenosyl-L-methionine reversible trans...	100.0%	Unknown
B_MANNASEly	beta-mannosidase, lysosomal	100.0%	Unknown
ILEt4	L-isoleucine transport in via sodium sym...	100.0%	Unknown
PCFLOPm	phosphatidylcholine flippase	100.0%	Unknown
PETOHMm_hs	phosphatidylethanolamine N-methyltransfe...	100.0%	Unknown
ADENOSYLHOMOCYSTEINASE_45_RXN	ADENOSYLHOMOCYSTEINASE-RXN	100.0%	Unknown

EXECUTIVE SUMMARY

This report presents a comprehensive metabolic modeling analysis of Human Metapneumovirus (HMPV) infection in human epithelial cells. The analysis involved construction of a Viral Biomass Objective Function (VBOF) and integration with the host metabolic model (iHsaEC21).

KEY RESULTS:

1. VBOF CONSTRUCTION

- Successfully constructed HMPV VBOF incorporating:
 - Genome nucleotides (13,350 bp)
 - 9 viral proteins with copy numbers
 - Lipid envelope composition
 - Glycan modifications (N-linked and O-linked)
 - Energy requirements

2. MODEL INTEGRATION

- Integrated VBOF into host model: iHsaEC21_CLEAN
- Baseline VBOF flux: 1.2654365695792906
- 38/42 metabolites successfully mapped

3. ANTIVIRAL TARGET IDENTIFICATION

- Tested 4660 genes
- Tested 3472 reactions
- Identified 10 essential gene targets
- Identified 26 essential reaction targets

KEY FINDINGS:

The analysis identified critical host metabolic pathways essential for HMPV replication, including:

- Sphingolipid biosynthesis (viral envelope formation)
- Phospholipid metabolism (membrane production)
- Glycan processing (glycoprotein maturation)
- Nucleotide metabolism (genome replication)
- Amino acid transport (protein synthesis)

These pathways represent potential targets for antiviral drug development.